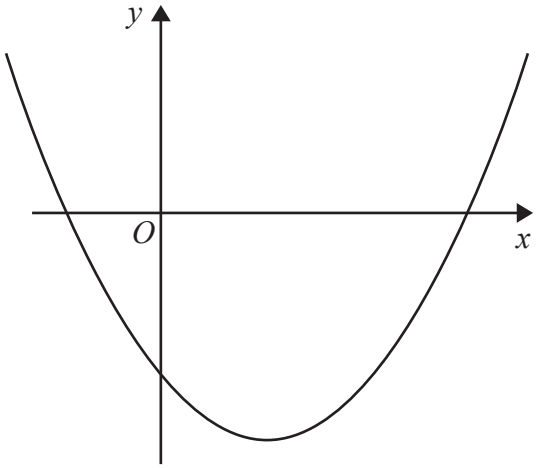
17 Here is a sketch of a curve.



The equation of the curve is $y = x^2 + ax + b$ where a and b are integers.

The points $(0, -5)$ and $(5, 0)$ lie on the curve.

Find the coordinates of the turning point of the curve.

At (0, −5): $-5 = 0 + 0 + b$, so $b = -5$.

1 mark
Substitute a given point

At (5, 0): $0 = 25 + 5a - 5$

$5a = -20$, so $a = -4$.

1 mark
Use the other point to find a

$y = x^2 - 4x - 5$; factors of −5 with sum −4: −5 and 1.

	x	−5
x	x^2	$-5x$
1	x	−5

$y = (x - 5)(x + 1)$: roots are 5 and −1.

Turning-point $x = \dfrac{5 + (-1)}{2} = 2$

1 mark
Midpoint of the roots

At $x = 2$: $y = 4 - 8 - 5 = -9$.

1 mark
Both coordinates found

(2 , −9)

(Total for Question 17 is 4 marks)

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